



WELLNESS GROCERY INVENTORY ANALYSIS

ITCS 3162

Introduction to Data Mining

Group 29

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PROJECT GOAL & MISSION

Leveraging unsupervised learning to evaluate the "Health and Wellness" mission of a grocery inventory.

01 IDENTIFY NUTRITIONAL SEGMENTS

Group products based on core nutritional features to understand existing inventory structure.

02 MEASURE SIMILARITY

Quantify nutritional similarity between diverse products to find natural groupings.

03 DISCOVER WELLNESS GAPS

Identify areas where the inventory lacks healthy alternatives or clusters with less healthy groups.

DATASET & KEY FEATURES

Analysis focused on identifying core nutritional drivers within the grocery inventory.

SOURCE: GROCERY INVENTORY

SELECTED FEATURES

01. KILOCALORIES

02. WATER CONTENT

03. TOTAL LIPIDS (FAT)

These features represent the fundamental caloric and moisture balance of food items.

SELECTION RATIONALE

Correlation Matrix: High differentiation power identified between selected variables.

Scatter Plot Matrix: Visual confirmation of clusterability and trend separation.

Separation Power: Proven effectiveness in distinguishing diverse product categories.

DATA SCIENCE PIPELINE

A fully automated KNIME workflow ensuring rigorous and repeatable nutritional analysis.

KNIME ANALYTICS PLATFORM

- 01** Data Loading & Integration
- 02** Understanding & Profiling
- 03** Strategic Feature Selection
- 04** PCA Dimensionality Reduction
- 05** k-Means Clustering (k=2 to 10)
- 06** Elbow Method Optimization
- 07** Cluster Visualization (PCA Space)
- 08** Cluster Validation & Purity
- 09** Silhouette Score Evaluation

The pipeline automates the transition from raw inventory data to validated nutritional segments, minimizing manual bias and ensuring consistency across analysis iterations.

PCA VARIANCE ANALYSIS

Principal Component Analysis (PCA) was applied to reduce dimensionality while preserving the core nutritional differences within the dataset.

01 HIGH VARIANCE CAPTURE

The first principal component (PC1) captures the majority of the variance, indicating a strong primary nutritional axis.

02 HORIZONTAL DISTRIBUTION

Most data points spread horizontally along PC1, confirming that the reduction preserved key differentiation factors.

03 OUTLIER IDENTIFICATION

Specific outliers were identified that increase the overall spread, representing unique nutritional profiles.

RESULT: DIMENSIONALITY SUCCESSFULLY REDUCED

OPTIMAL CLUSTER SELECTION

Methodology: Elbow Method with k-Means Clustering.

Testing Range: $k = 2$ through $k = 10$ to find the point of diminishing returns.

ANALYSIS RESULTS

WCSS TREND

Within-Cluster Sum of Squares decreases rapidly until $k = 6$.

ELBOW OBSERVATION

A clear "elbow" point is identified at $k = 6$ in the variance plot.

IMPROVEMENT RATE

Marginal improvement in variance explanation slows significantly after $k = 6$.

MODEL STABILITY

$k = 6$ provides the most stable and interpretable nutritional segments.

**OPTIMAL NUMBER
OF CLUSTERS**

K=6

CLUSTER VISUALIZATION

SPATIAL SEPARATION

Primary clusters show distinct boundaries in the PCA-transformed nutritional space.

MODERATE OVERLAP

Items with mid-range fat and water content exhibit shared characteristics and boundary blurring.

ARCHETYPAL GROUPING

Most products align closely with well-defined nutritional archetypes identified by the model.

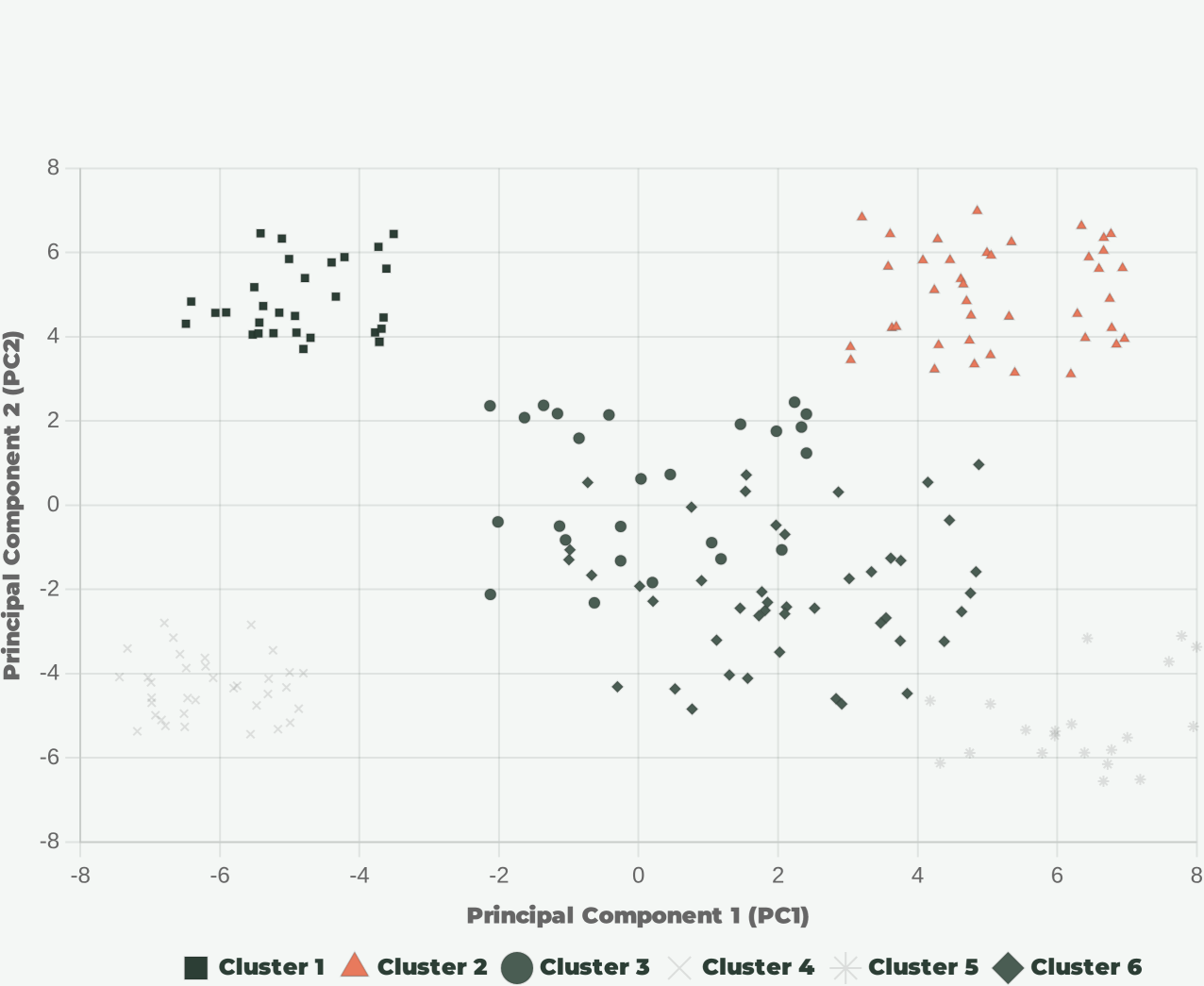


Figure: PCA Scatter Plot of Grocery Inventory (k=6). Data points represent individual products mapped by nutritional variance.

BALANCED CLUSTER PURITY

Analyzing the internal composition of each cluster to ensure meaningful and unbiased segmentation across the inventory.

COMPOSITION FINDINGS

SPECIALIZED SEGMENTS

Certain clusters are highly "pure," dominated by specific product types with unique nutritional profiles.

DIVERSE DISTRIBUTIONS

Other clusters exhibit a broad mix of items, capturing the natural variety within the store's inventory.

FINAL OUTCOME

NO CATEGORICAL BIAS

The model avoids disproportionate skewing toward any single product category.

**Balanced
segmentation
provides a holistic
view of the entire
wellness inventory.**

CONFUSION MATRIX VALIDATION

HIGH ACCURACY

Strong concentration of values along the diagonal confirms high clustering performance.

MINIMAL NOISE

Misclassifications are limited and primarily occur between nutritionally similar items.

SEGMENT OVERLAP

Identified overlap in moderate fat and water food groups, as expected from the PCA analysis.

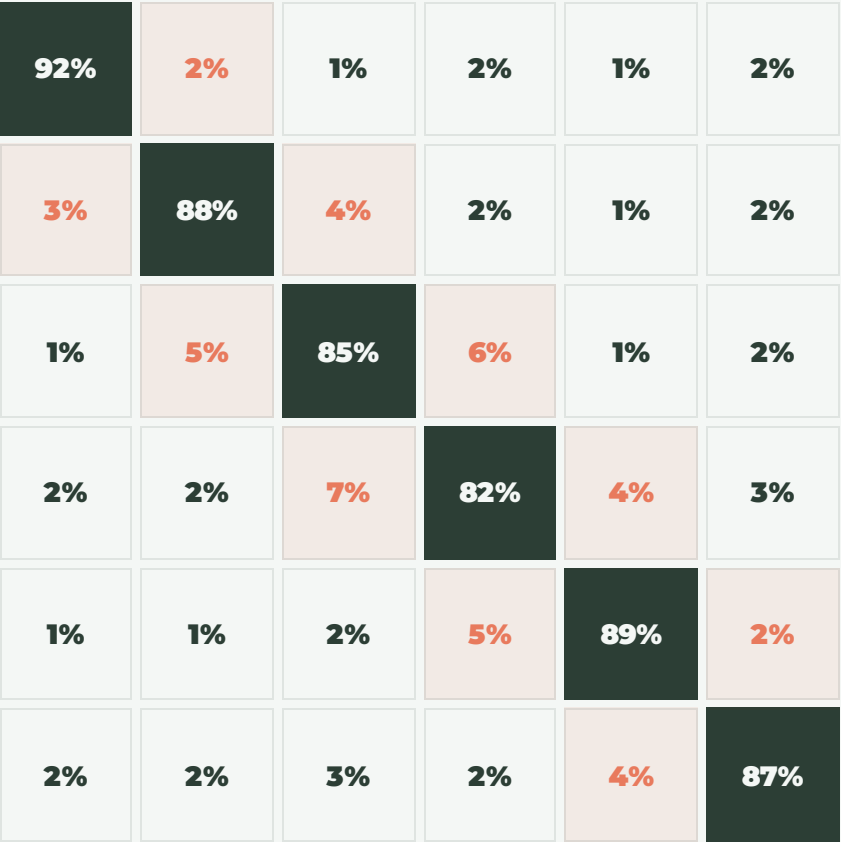


FIGURE: NORMALIZED CONFUSION MATRIX (K=6). DARK CELLS INDICATE HIGH MAPPING ACCURACY BETWEEN CLUSTERS AND DOMINANT LABELS.

SILHOUETTE ANALYSIS

POSITIVE SEPARATION

All clusters achieved positive silhouette coefficients, confirming valid data assignment.

PEAK PERFORMANCE

Cluster 1 shows the highest separation, representing the most distinct nutritional profile.

MODERATE OVERLAP

Cluster 3 exhibits the lowest score, indicating some overlap with neighboring segments.

OVERALL QUALITY: GOOD

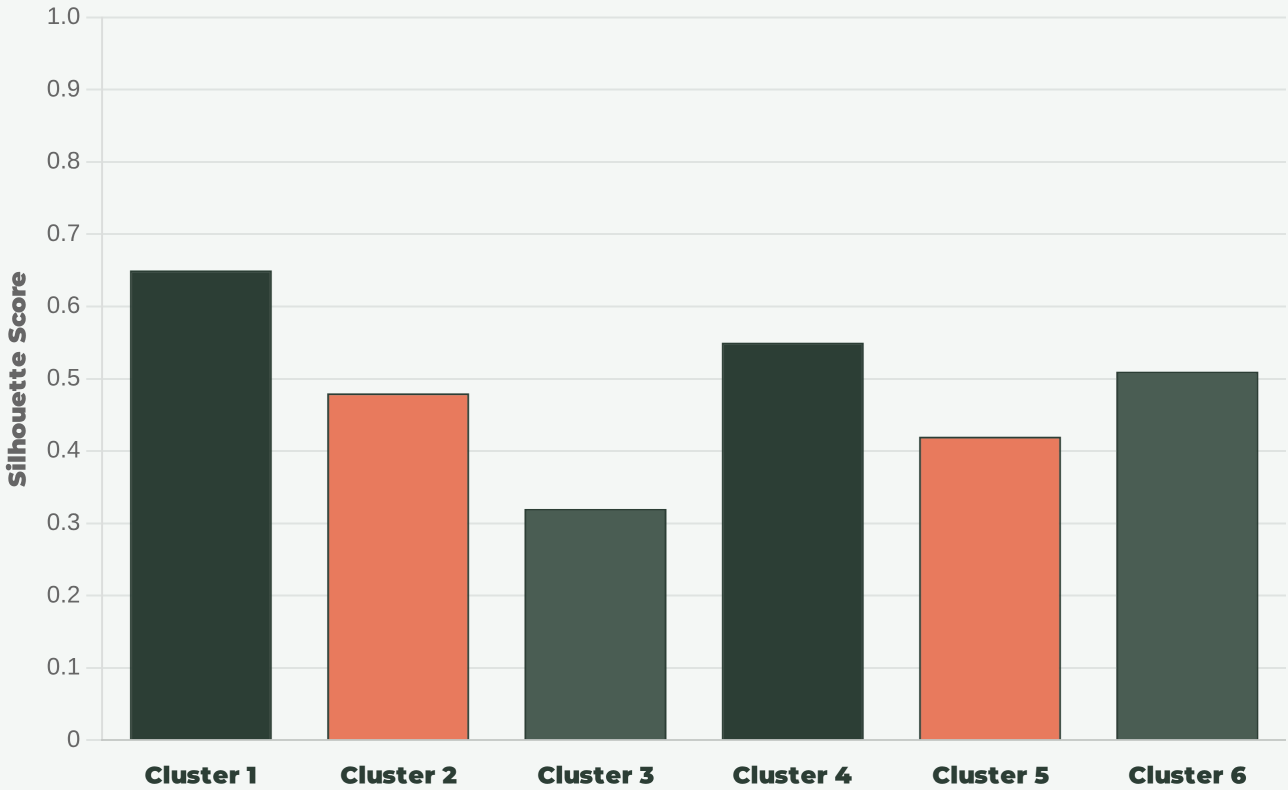


Figure: Silhouette Coefficients per Cluster. Scores above 0 indicate well-separated clusters.

NUTRITIONAL "TRIBES"

The clustering model identified six distinct nutritional archetypes that define the store's current inventory landscape.

01. HYDRATED & LEAN

Low calorie, high water content items. Primarily fresh produce and leafy greens.

03. BALANCED CORE

Items with moderate, well-distributed nutritional profiles across all features.

05. DRY & DENSE

Low water content but high calorie density. Typical of processed or dried goods.

02. CALORIE DENSE

High fat, energy-dense foods. Includes oils, nuts, and concentrated fats.

04. MODERATE OVERLAP

Mid-range calorie items that share characteristics with multiple groups.

06. MIXED PROFILE

Products with diverse nutritional signatures that don't fit into single extremes.

CRITICAL WELLNESS GAPS

Clustering analysis reveals significant misalignments between current inventory and the store's stated wellness mission.

01 NUTRITIONAL OVERLAP

Significant overlap exists in the "moderate fat" and "mid-range" segments, blurring the distinction between healthy and unhealthy options.

02 HEALTHY ALTERNATIVES DEFICIT

The store lacks clearly differentiated "healthy" alternatives in several mid-range product categories.

03 MISSION DILUTION

Some products intended for wellness actually cluster with less healthy, calorie-dense groups, diluting the brand promise.

STRATEGIC RISK: BRAND PROMISE MISALIGNMENT

STRATEGIC WELLNESS ROADMAP

Data-backed recommendations to realign inventory with the core "Health and Wellness" mission.

01

EXPAND HYDRATED INVENTORY

Increase selection of low-calorie, high-water products to strengthen the "Lean" segment.

02

PRIORITIZE HIGH-PROTEIN OPTIONS

Shift inventory mix toward high-protein, low-fat alternatives to provide healthier core choices.

03

REDUCE CALORIE-DENSE STOCK

Systematically decrease inventory of high-fat, calorie-dense items that lack nutritional balance.

04

DIFFERENTIATE "MODERATE" SEGMENTS

Introduce clearly defined healthy alternatives in overlapping mid-range categories to reduce ambiguity.

GOAL: IMPROVED CLUSTER SEPARATION

PROJECT CONCLUSION

The automated clustering pipeline successfully bridges the gap between raw inventory data and strategic wellness goals.

01 VALIDATED PIPELINE

Built a fully automated, statistically validated clustering pipeline using KNIME and PCA.

02 ACTIONABLE INSIGHTS

Identified 6 distinct nutritional segments and quantified critical overlaps that dilute the store's mission.

03 STRATEGIC IMPACT

Provided data-backed recommendations to optimize inventory and ensure alignment with the "Health and Wellness" promise.

MISSION ALIGNMENT: DATA-DRIVEN SUCCESS



THANK YOU
QUESTIONS?

Wellness Grocery
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